

The Dyadic Web of the Fabius Function

Moment corrections, half-base q -interpolation, binary telescopes, and discrete extrapolation

A complementary exposition to *The Fabius Function and Rvachev's Up-Function*
based on Vladimir Reshetnikov's ProveIt formal development

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Abstract

Exact values of the Fabius function at dyadic rationals appear in several forms that look unrelated: a Thue–Morse sum corrected by the even moments of Rvachev's up-function, a finite half-base q -binomial expression with a seemingly arbitrary complex translation, a bit-recursive Taylor reduction, an absolutely convergent binary series at arbitrary argument, and a finite q -weighted discrete approximant whose limit is the signed global Fabius function. This companion article identifies the common algebra behind these formulae and makes their connections explicit.

The central object is the coefficient

$$[X^n](B_m(X)\mathcal{A}(X)), \quad B_m(X) = \sum_{h < m} (-1)^{\text{wt}_2(h)} e^{(m-1-h)X},$$

where $\mathcal{A}$ is the exponential moment series of the Fabius probability law. Direct expansion gives the classical moment formula; probabilistic interpretation gives an expectation formula; Cauchy's formula gives a contour representation; and geometric divided differences through $-1, -2, \dots, -2^n$ give the q -binomial formula. The same coefficient also produces new finite identities: an exact moment-corrected expansion in centered splines, its reciprocal deconvolution, and a residue formula for the Gaussian interpolation row.

At arbitrary arguments the two infinite constructions have different architectures. The binary series is a genuine telescoping sum whose summands are individually independent of the complex shift. The discrete formula is instead a Toeplitz transform of shifted splines; its finite rows generally depend on the shift, although their limit does not. Writing the row generating polynomial as

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}$$

reveals it as a q -Richardson filter: it preserves constants and annihilates the geometric error modes $2^{-p}, 2^{-2p}, \dots, 2^{-np}$. This yields an explicit uniform $\mathcal{O}(4^{-n})$ error bound. Most importantly, at a dyadic input $m/2^s$ the discrete approximant stabilizes exactly for every row $n \geq s$; thus the “limit formula” literally becomes the exact dyadic q -formula after the denominator has been resolved. A synchronized nearest-dyadic version provides a second, termwise translation-invariant discrete limit.

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1 Scope, conventions, and the formula map

This article is a companion rather than a replacement for the comprehensive exposition [12]. It assumes the construction and basic analytic theory developed there and concentrates on one cluster of results: exact dyadic evaluation, binary reduction, finite half-base q -calculus, and discrete limits. The proofs formalized in Lean under `Analysis/FabiusFunction`[11] are the source of the exact identities used below. Several consequences in this article are then derived directly from those identities; a formalization map is given in section 9.

We use three related functions, with the same convention as the companion article:

- F is the bounded Fabius distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported Rvachev density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on $\mathbb{R}$.

They agree on $[0, 1]$. Put

$$\varepsilon_r = (-1)^{\text{wt}_2(r)}, \quad c_k = \int_{-1}^1 u^{2k} \text{up}(u) \, du,$$

and let d_n denote the half moments normalized by

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \quad (1)$$

The normalized moment series is

$$\mathcal{A}(X) = \sum_{n \geq 0} d_n \frac{X^n}{n!}. \quad (2)$$

We reserve q for the fixed base $q = 1/2$ and use $Q \in \mathbb{C}$ for a freely chosen translation. This distinction is essential: in the expressions $(a; q)_n$ and $(r - 2^k + Q)^{n+k}$ the two symbols play completely different roles.

1.1 Five presentations of the same value

The identities studied below fall into five families. Their superficial differences are summarized in table 1; the rest of the article explains the arrows between rows.

The main structural conclusions can be stated before any calculation.

The common core. The exact dyadic formulae are all coefficient extractors applied to the same entire function $B_m \mathcal{A}$. The binary series repeatedly reuses its Taylor coefficients at successive binary scales. The discrete formula applies a normalized q -Pochhammer polynomial to the sequence of finite-spline degrees. At a dyadic input these two arbitrary-argument mechanisms both become finite: the binary series terminates, while the discrete rows stabilize exactly.

1.2 The exact dyadic formula taken as input

For $0 \leq m \leq 2^n$ the moment formula proved in the formal development is

$$F\left(\frac{m}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2m - 2h - 1)^{n-2k}. \quad (3)$$

Presentation	Range and status	Principal coefficients	Mechanism
Moment/Thue–Morse formula	finite and exact at $m/2^n$	c_k and ε_h	coefficient expansion in the centered probability law
Half-base q -formula	finite and exact at $m/2^n$	$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ and $(1/2; 1/2)_n$	geometric divided difference after block cancellation
Binary Taylor series	infinite and absolutely convergent for $x \geq 0$; finite at dyadics	binary prefixes and d_j	exact residual telescope along the binary expansion
Global q -binary series	same outer series as the preceding row	a finite q -identity inside each scale	termwise replacement of a Taylor polynomial
Discrete q -limit	one finite row for each n , convergent for all $x \in \mathbb{R}$	normalized Gaussian row $\omega_{n,j}$	Toeplitz transform of shifted finite splines

Table 1: The formula families. The two infinite constructions are not partial-sum versions of one another.

The same finite expression, interpreted with $\mathcal{F}$ on the left, is valid for every $m, n \in \mathbb{N}$. The q -binomial version is

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} \varepsilon_r(r - m2^k + Q)^{n+k}, \end{aligned} \quad (4)$$

valid for every $Q \in \mathbb{C}$ and independent of Q as a complete finite expression. A central goal below is to explain why (3) and (4) are two readings of one coefficient.

2 The master coefficient and its probabilistic meaning

2.1 The probability law behind the moment series

Let $U_1, U_2, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$, and set

$$Y = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad Z = 2Y - 1. \quad (5)$$

Then $Y \in [0, 1]$, its distribution function is F , and Z has density up on $[-1, 1]$. Consequently

$$d_n = \mathbb{E}(Y^n), \quad c_k = \mathbb{E}(Z^{2k}), \quad \mathbb{E}(Z^{2k+1}) = 0. \quad (6)$$

The moment generating function of one uniform variable is

$$E_0(X) = \frac{e^X - 1}{X} = \int_0^1 e^{Xu} du,$$

with the removable value $E_0(0) = 1$. Independence gives the locally uniformly convergent entire product

$$\mathcal{A}(X) = \mathbb{E}(e^{XY}) = \prod_{j=1}^{\infty} E_0(X/2^j). \quad (7)$$

Separating the first factor, or merely replacing X by $2X$ in the product, gives the refinement equation

$$\mathcal{A}(2X) = E_0(X)\mathcal{A}(X). \quad (8)$$

The centered moment series

$$\mathcal{C}(T) = \mathbb{E}(e^{TZ}) = \sum_{k \geq 0} c_k \frac{T^{2k}}{(2k)!} \quad (9)$$

satisfies the exact change of coordinates

$$\boxed{\mathcal{A}(X) = e^{X/2} \mathcal{C}(X/2)}. \quad (10)$$

The familiar hyperbolic-sine form follows by writing $E_0(t) = e^{t/2} \sinh(t/2)/(t/2)$:

$$\mathcal{A}(X) = e^{X/2} \prod_{j=1}^{\infty} \frac{\sinh(X/2^{j+1})}{X/2^{j+1}}, \quad (11)$$

$$\mathcal{C}(T) = \prod_{j=1}^{\infty} \frac{\sinh(T/2^j)}{T/2^j}. \quad (12)$$

Thus the moment series used in exact arithmetic is the real-exponential counterpart of the sinc product in Fourier analysis.

2.2 The Thue–Morse prefix series

For $m \in \mathbb{N}$ define

$$B_m(X) = \sum_{h=0}^{m-1} \varepsilon_h e^{(m-1-h)X}, \quad B_0(X) = 0. \quad (13)$$

This finite exponential polynomial records the numerator m and nothing else. Its two factors have distinct roles:

- B_m carries the finite Thue–Morse prefix and therefore the binary numerator;
- $\mathcal{A}$ carries the universal Fabius moments and therefore the analytic law.

Their product is the promised master object.

Theorem 2.1 (Master coefficient identity). *For every $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}\left(\frac{m}{2^n}\right) = 2^{-\binom{n}{2}} [X^n] (B_m(X) \mathcal{A}(X))}. \quad (14)$$

If $0 \leq m \leq 2^n$, the left side is the bounded value $F(m/2^n)$.

Proof. Put $t_h = 2m - 2h - 1$. By (9),

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k t_h^{n-2k} = n! [W^n] (e^{t_h W} \mathcal{C}(W)).$$

Using (10) at $X = 2W$ gives $\mathcal{C}(W) = e^{-W} \mathcal{A}(2W)$. Since $t_h - 1 = 2(m - 1 - h)$,

$$\begin{aligned} n! [W^n] (e^{t_h W} \mathcal{C}(W)) &= n! [W^n] (e^{2(m-1-h)W} \mathcal{A}(2W)) \\ &= n! 2^n [X^n] (e^{(m-1-h)X} \mathcal{A}(X)). \end{aligned}$$

Multiply by ε_h , sum over $h < m$, and insert the result in (3). The power of two becomes $2^{n - \binom{n+1}{2}} = 2^{-\binom{n}{2}}$. $\square$

The theorem compresses the double sum in (3) into one coefficient, but it also expands naturally in several different directions.

Corollary 2.2 (Expectation form). *For every $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \mathbb{E}(m-1-h+Y)^n \quad (15)$$

$$= \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \mathbb{E}(2m-2h-1+Z)^n. \quad (16)$$

Proof. Because $\mathcal{A}(X) = \mathbb{E}(e^{XY})$,

$$[X^n](e^{aX} \mathcal{A}(X)) = \frac{1}{n!} \mathbb{E}(a+Y)^n.$$

Apply this termwise to B_m . The second form follows from $m-1-h+Y = (2m-2h-1+Z)/2$. $\square$

Expanding the centered expectation in (16), using symmetry to discard the odd moments, recovers (3) immediately. In this sense the Thue–Morse/moment formula is simply the binomial theorem applied inside one probability expectation.

Corollary 2.3 (Cauchy coefficient representation). *For any $\rho > 0$,*

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{2\pi i} \int_{|X|=\rho} \frac{B_m(X) \mathcal{A}(X)}{X^{n+1}} dX. \quad (17)$$

This contour formula is elementary, but useful conceptually: the q -binomial formula will turn out to be a second residue computation, now in an auxiliary geometric node variable.

2.3 The numerator and the law separate completely

Expanding (14) without centering gives another exact finite form:

$$\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^n \binom{n}{j} d_j (m-1-h)^{n-j}. \quad (18)$$

The d_j are universal and may be precomputed once. The numerator enters only through the finite power sums $\sum_{h < m} \varepsilon_h (m-1-h)^r$. This is often a more transparent bridge to the Taylor reduction series than the centered c_k formula, because the reduction polynomials themselves are written in the d_j .

Remark 2.4 (Two independent cancellations). The Thue–Morse prefix in (18) may annihilate low ordinary powers when m is a complete binary block. The q -binomial formula introduces a second cancellation across the geometric scale index. These cancellations live in different variables and should not be conflated: one acts on h or r inside a binary block, the other on the scale node -2^k .

3 Moment coordinates and alternative exact representations

3.1 The triangular transforms between c_k and d_n

The affine relation $Z = 2Y - 1$ gives both directions of the moment change of coordinates.

Proposition 3.1 (Moment transforms). *For every $n, k \in \mathbb{N}$,*

$$d_n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k, \quad (19)$$

$$c_k = \sum_{j=0}^{2k} (-1)^j 2^j \binom{2k}{j} d_j. \quad (20)$$

Proof. Since $Y = (1 + Z)/2$, expand Y^n and use the vanishing odd moments of Z . Conversely expand $(2Y - 1)^{2k}$; because $2k$ is even, the sign of the term of degree j is $(-1)^j$. $\square$

The first formula explains the odd-looking shift $2m - 2h - 1$ in (3): it is precisely the centered coordinate obtained after converting the half moments to the even moments of up.

3.2 An infinite-cube composition formula

Each factor in (7) has the expansion

$$E_0(X/2^j) = \sum_{r \geq 0} \frac{X^r}{2^{jr} (r+1)!}.$$

Taking the coefficient of X^n gives an exact positive series over finite-support weak compositions.

Proposition 3.2 (Composition representation of the half moments). *For every $n \in \mathbb{N}$,*

$$d_n = n! \sum_{\substack{r_1, r_2, \dots \geq 0 \\ r_1 + r_2 + \dots = n}} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j + 1)!}. \quad (21)$$

Only finitely many r_j are nonzero in each summand, and the series converges absolutely. Equivalently,

$$d_n = \lim_{J \rightarrow \infty} \int_{[0,1]^J} \left(\sum_{j=1}^J \frac{u_j}{2^j} \right)^n du_1 \cdots du_J. \quad (22)$$

This gives a q -free, manifestly positive representation of every inverse-dyadic value:

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} d_n = 2^{-\binom{n}{2}} \sum_{r_1 + r_2 + \dots = n} \prod_{j \geq 1} \frac{2^{-jr_j}}{(r_j + 1)!}. \quad (23)$$

3.3 Cumulants, Bernoulli numbers, and Bell polynomials

The infinite product also gives a compact partition formula. With Bernoulli numbers in the convention $B_2 = 1/6$, $B_4 = -1/30$, one has

$$\log \frac{e^X - 1}{X} = \frac{X}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} X^{2r}. \quad (24)$$

Summing this identity over the dyadic scales in (7) yields

$$\log \mathcal{A}(X) = \frac{X}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!(2^{2r} - 1)} X^{2r}. \quad (25)$$

Thus the cumulants κ_j of Y are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0 \quad (r \geq 1). \quad (26)$$

Theorem 3.3 (Bell-polynomial representation). *Let $\mathcal{B}_n$ be the complete exponential Bell polynomial. Then*

$$d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (27)$$

and explicitly

$$d_n = n! \sum_{\substack{m_1, \dots, m_n \geq 0 \\ 1m_1 + 2m_2 + \dots + nm_n = n}} \prod_{r=1}^n \frac{1}{m_r!} \left(\frac{\kappa_r}{r!} \right)^{m_r}. \quad (28)$$

Consequently

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (29)$$

Proof. Equation (25) is the standard cumulant expansion $\log \mathcal{A}(X) = \sum_{j \geq 1} \kappa_j X^j / j!$. Exponentiating and comparing coefficients is exactly the defining generating identity for the complete Bell polynomials. $\square$

The representation is useful when Bernoulli arithmetic is already available. It also gives the moment recurrence

$$d_{n+1} = \sum_{j=0}^n \binom{n}{j} \kappa_{j+1} d_{n-j}, \quad (30)$$

while (8) gives the rational recurrence

$$(2^n - 1)d_n = \sum_{r=1}^n \binom{n}{r} \frac{d_{n-r}}{r+1} \quad (n \geq 1). \quad (31)$$

n	d_n	c_n	$F(2^{-n})$
0	1	1	1
1	1/2	1/9	1/2
2	5/18	19/675	5/72
3	1/6	583/59535	1/288
4	143/1350	132809/32531625	143/2073600
5	19/270	46840699/24405225075	19/33177600

Table 2: The columns use different index conventions: c_n is the $2n$ th centered moment, whereas d_n is the n th half moment.

3.4 Exact moment correction of the matched spline

The centered finite spline of degree $p \geq 1$ is

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2} \right)^p, \quad N_p(x) = \max \left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor \right). \quad (32)$$

At degree zero set

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r. \quad (33)$$

The companion article proves the global estimate

$$\sup_{x \in \mathbb{R}} |S_p(x) - \mathcal{F}(x)| \leq 2^{-p}. \quad (34)$$

At the matched grid point $m/2^p$ the cutoff is m and

$$S_p(m/2^p) = \frac{1}{2^{\binom{p+1}{2}} p!} \sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p. \quad (35)$$

The $k = 0$ part of (3) is exactly this matched spline. The remaining moments supply a finite correction hierarchy.

Theorem 3.4 (Moment-corrected matched-spline identity). *For all $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)! 2^{k(2n-2k+1)}} S_{n-2k} \left(\frac{m}{2^{n-2k}} \right). \quad (36)$$

For $0 \leq m \leq 2^n$ the left side is $F(m/2^n)$. The spline arguments on the right are understood globally and may lie outside $[0, 1]$.

Proof. In (3) put $p = n - 2k$. By (35),

$$\sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p = 2^{\binom{p+1}{2}} p! S_p(m/2^p).$$

Moreover

$$\frac{1}{n!} \binom{n}{2k} p! = \frac{1}{(2k)!},$$

and

$$\binom{n+1}{2} - \binom{p+1}{2} = k(2n - 2k + 1).$$

Substitution gives (36). □

The formula has three noteworthy features.

1. Its first term is the obvious finite-spline approximation $S_n(m/2^n)$.
2. The correction drops the spline degree by two at each step, reflecting the evenness of up and the disappearance of odd centered moments.
3. The numerator m remains fixed, while the denominator is coarsened from 2^n to 2^{n-2k} . Thus the formula couples degree reduction to dyadic rescaling.

Corollary 3.5 (Exact defect of the matched spline). *For every $m, n \in \mathbb{N}$,*

$$\mathcal{F}(m/2^n) - S_n(m/2^n) = \sum_{k=1}^{\lfloor n/2 \rfloor} \frac{c_k}{(2k)! 2^{k(2n-2k+1)}} S_{n-2k} \left(\frac{m}{2^{n-2k}} \right). \quad (37)$$

This is stronger than an error estimate: it identifies the error exactly as a finite sequence of lower-degree global splines. The small factor in the first correction is $c_1/(2 \cdot 2^{2n-1}) = 1/(9 \cdot 2^{2n})$.

3.5 The reciprocal correction transform

The previous relation is triangular and can be inverted. Define reciprocal centered moments γ_k by

$$\frac{1}{\mathcal{C}(T)} = \sum_{k \geq 0} \gamma_k \frac{T^{2k}}{(2k)!}. \quad (38)$$

Thus

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{k=0}^{n-1} \binom{2n}{2k} \gamma_k c_{n-k} \quad (n \geq 1), \quad (39)$$

and the first values are

$$1, \quad -\frac{1}{9}, \quad \frac{31}{675}, \quad -\frac{2347}{59535}, \quad \frac{1904369}{32531625}, \dots \quad (40)$$

Theorem 3.6 (Reciprocal moment deconvolution). *For every $m, n \in \mathbb{N}$,*

$$S_n(m/2^n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_k}{(2k)! 2^{k(2n-2k+1)}} \mathcal{F}\left(\frac{m}{2^{n-2k}}\right). \quad (41)$$

Proof. For a fixed numerator m , introduce

$$P_p(m) = \sum_{h=0}^{m-1} \varepsilon_h (2m - 2h - 1)^p, \quad G_p(m) = 2^{\binom{p+1}{2}} p! \mathcal{F}(m/2^p).$$

The exact moment formula says, in exponential generating functions,

$$\sum_{p \geq 0} G_p(m) \frac{T^p}{p!} = \mathcal{C}(T) \sum_{p \geq 0} P_p(m) \frac{T^p}{p!}.$$

Multiply by $1/\mathcal{C}(T)$, compare coefficients, and then use $P_p(m) = 2^{\binom{p+1}{2}} p! S_p(m/2^p)$. The same triangular exponent simplification as in [theorem 3.4](#) gives (41). $\square$

Equations (36) and (41) are a finite convolution/deconvolution pair. They show that the exact dyadic values and the matched centered splines carry precisely the same information once the universal moment series $\mathcal{C}$ is known.

First bridge to the discrete theory. The centered spline is not merely an approximation from which exact dyadic arithmetic is separate. At a matched grid point, the exact value is obtained by a finite universal moment correction of that spline, and the correction is invertible. Later, the half-base q -Pochhammer row will give a second exact correction, using consecutive spline degrees at the *same* argument rather than parity-separated degrees at rescaled arguments.

4 The half-base q -formula as geometric coefficient extraction

4.1 The Gaussian row is a divided difference

For a general base q , write

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (42)$$

and let $\begin{bmatrix} n \\ k \end{bmatrix}_q$ denote the Gaussian binomial coefficient. At $q = 1/2$ the finite q -binomial theorem is

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \quad (43)$$

Put

$$x_k = -2^k, \quad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2}, \quad \mathfrak{m}_n = \prod_{\ell=1}^n (2^\ell - 1). \quad (44)$$

Then

$$\sum_{k=0}^n w_{n,k} x_k^d = \begin{cases} 0, & 0 \leq d < n, \\ \mathfrak{m}_n, & d = n, \end{cases} \quad (45)$$

and

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{\ell \neq k} (x_k - x_\ell)}, \quad (1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \quad (46)$$

Thus the functional

$$\mathfrak{D}_n[P] = \frac{1}{\mathfrak{m}_n} \sum_{k=0}^n w_{n,k} P(-2^k) \quad (47)$$

is the divided difference $[-1, -2, \dots, -2^n]P$. For every polynomial of degree at most n , it returns the leading coefficient.

Proposition 4.1 (Residue form of the geometric extractor). *Let*

$$\Pi_n(z) = \prod_{k=0}^n (z + 2^k),$$

and let P be a polynomial of degree at most n . If Γ encloses all nodes $-1, -2, \dots, -2^n$, then

$$[z^n]P(z) = \mathfrak{D}_n[P] = \frac{1}{2\pi i} \int_{\Gamma} \frac{P(z)}{\Pi_n(z)} dz. \quad (48)$$

Proof. The residues at the simple poles $x_k = -2^k$ are $P(x_k)/\Pi'_n(x_k) = P(x_k)/\prod_{\ell \neq k} (x_k - x_\ell)$, giving the barycentric sum in (47). Lagrange interpolation identifies that sum with the leading coefficient. $\square$

The q -symbols in the dyadic formula therefore have a concrete and limited task: they are the closed form of a divided-difference stencil on a geometric grid. No infinite basic-hypergeometric summation is involved.

4.2 The parametric coefficient polynomial

Introduce an auxiliary node variable z and define

$$\mathcal{R}_m(z; X) = \frac{e^{-X} B_m(zX) \mathcal{A}(zX)}{\mathcal{A}(-X)}, \quad r_{m,n}(z) = [X^n] \mathcal{R}_m(z; X). \quad (49)$$

Because $\mathcal{A}(-X)$ is a unit with constant term 1, this is a well-defined formal series and an entire analytic germ. More importantly, $r_{m,n}$ is a polynomial in z .

Lemma 4.2 (Node polynomial and its leading coefficient). *For fixed m, n , the polynomial $r_{m,n}(z)$ has degree at most n and*

$$[z^n] r_{m,n}(z) = [X^n] (B_m(X) \mathcal{A}(X)) = 2^{\binom{n}{2}} \mathcal{F}(m/2^n). \quad (50)$$

Proof. Write $e^{-X}/\mathcal{A}(-X) = \sum_{j \geq 0} u_j X^j$ with $u_0 = 1$. Since $B_m(zX) \mathcal{A}(zX) = (B_m \mathcal{A})(zX)$, the coefficient of X^n is a polynomial in z of degree at most n . Its z^n term can only use the constant u_0 , so its leading coefficient is $[X^n](B_m \mathcal{A})$. Apply [theorem 2.1](#). $\square$

The exact dyadic value is now the leading coefficient of a degree- n polynomial sampled at $-1, -2, \dots, -2^n$. Applying [theorem 4.1](#) gives a compact alternative to the displayed q -sum.

Corollary 4.3 (Geometric-node contour representation). *For every $m, n \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(m/2^n) = \frac{2^{-\binom{n}{2}}}{2\pi i} \int_{\Gamma} \frac{r_{m,n}(z)}{\prod_{k=0}^n (z + 2^k)} dz,} \quad (51)$$

where Γ encloses the geometric nodes.

Together, (17) and (51) exhibit two independent Cauchy extractions:

coefficient in the analytic variable X $[X^n](B_m \mathcal{A})$ ordinary Cauchy kernel X^{-n-1}	leading coefficient in the geometric node z $[z^n]r_{m,n}(z)$ barycentric kernel $\Pi_n(z)^{-1}$.
---	--

The finite q -binomial formula is what results when the second contour is evaluated by residues and each node value is converted to a Thue–Morse block sum.

4.3 Why the node values are Thue–Morse power sums

For a complete binary block set

$$C_k(X) = \sum_{r=0}^{2^k-1} \varepsilon_r e^{(r-2^k)X}. \quad (52)$$

The digit factorization $\sum_{r < 2^k} \varepsilon_r z^r = \prod_{j=0}^{k-1} (1 - z^{2^j})$ gives

$$C_k(X) = (-1)^k 2^{\binom{k}{2}} X^k e^{-X} \prod_{j=0}^{k-1} E_0(-2^j X). \quad (53)$$

Hence C_k has a zero of exact order k . The refinement (8), iterated k times, gives

$$\frac{e^{-X} \mathcal{A}(-2^k X)}{\mathcal{A}(-X)} = e^{-X} \prod_{j=0}^{k-1} E_0(-2^j X). \quad (54)$$

Combining (53) and (54) identifies the node specialization of (49) with a normalized complete block.

For a general numerator, the long block factors as

$$\sum_{r=0}^{m2^k-1} \varepsilon_r e^{(r-m2^k)X} = B_m(-2^k X) C_k(X). \quad (55)$$

After division by the forced factor $(-1)^k 2^{\binom{k}{2}} X^k$, the coefficient of X^n is

$$\frac{(-1)^k 2^{-\binom{k}{2}}}{(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k)^{n+k}. \quad (56)$$

Multiplication by the interpolation weight $w_{n,k}$ cancels the sign and contributes a second factor $2^{-\binom{k}{2}}$. Their product is the characteristic $4^{-\binom{k}{2}}$ of (4). Finally (46) converts m_n into $2^{n^2} (1/2; 1/2)_n$ after the factor $2^{\binom{n}{2}}$ from [theorem 4.2](#) is included. This recovers the exact q -formula with $Q = 0$.

4.4 Translation invariance is lower-degree annihilation

The shift Q enters a block generating function by multiplication with e^{QX} . If $P_{m,k}(X)$ denotes the normalized long-block series after its forced X^k has been removed, then the q -numerator is a weighted sum of $[X^n](e^{QX}P_{m,k}(X))$. Expanding the exponential,

$$[X^n](e^{QX}P_{m,k}(X)) = \sum_{s=0}^n \frac{Q^{n-s}}{(n-s)!} [X^s]P_{m,k}(X). \quad (57)$$

For each $s < n$, the Gaussian row sees a polynomial of geometric-node degree at most s and annihilates it. Only the $s = n$ term survives; it has no Q factor.

Proposition 4.4 (Finite translation invariance; Lean formalized). *For fixed m, n , the polynomial*

$$Q \mapsto \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r(r - m2^k + Q)^{n+k} \quad (58)$$

is constant.

This is stronger than saying that a limit is independent of Q : every coefficient of $Q, Q^2, \dots$ vanishes in the finite expression itself. The distinction becomes important in [section 6](#), where the finite discrete rows generally do depend on Q . The constant-polynomial identity and its coefficientwise form are the exact declarations `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_l_eq_const` and `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_coeff` in `FabijsDyadicQBinomialScalar.lean`.

4.5 A useful abstract summary

The exact q -formula is a composition of two annihilators:

1. the complete Thue–Morse block has a zero of order k , removing all inner polynomial degrees below k ;
2. the Gaussian row on -2^k removes all remaining node degrees below n and returns the leading degree n .

The exponent $n+k$ in the inner power is therefore forced: k degrees are consumed by the binary block, leaving exactly n degrees for the geometric extractor. The formula is a hybrid of automatic-sequence cancellation, ordinary exponential generating functions, and finite q -calculus; it is not a terminating basic-hypergeometric summation in the usual ${}_r\phi_s$ sense [7, 9].

5 Binary Taylor reduction and the global series

5.1 The finite reduction polynomial

The inverse-dyadic moments assemble into the polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^m 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \quad (59)$$

The signed global extension satisfies the exact Taylor-block identity

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \quad 0 \leq y < 2^{-m}. \quad (60)$$

This is the elementary step behind both the terminating dyadic recursion and the infinite binary telescope.

For $x \geq 0$ set

$$p_m(x) = \lfloor 2^m x \rfloor, \quad y_m(x) = x - \frac{p_m(x)}{2^m}, \quad 0 \leq y_m(x) < 2^{-m}, \quad (61)$$

and put

$$p_{m-1}^*(x) = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}(x), & m \geq 1. \end{cases} \quad (62)$$

Define

$$\alpha_m(x) = (-1)^{\text{wt}_2(p_m(x))} (2p_{m-1}^*(x) - p_m(x)). \quad (63)$$

This coefficient is 0 unless the newly exposed binary digit is 1, in which case it is a sign.

Theorem 5.1 (Binary reduction series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{T}_m(y_m(x)).} \quad (64)$$

The series converges absolutely, and its partial sum through $m = N$ has the uniform remainder bound

$$\left| \mathcal{F}(x) - \sum_{m=0}^N \alpha_m(x) \mathcal{T}_m(y_m(x)) \right| \leq 2^{1-N} \quad (N \geq 1). \quad (65)$$

The proof is an exact residual telescope. If $R_m(x) = (-1)^{\text{wt}_2(p_m(x))} \mathcal{F}(y_m(x))$, then

$$\alpha_m(x) \mathcal{T}_m(y_m(x)) = R_{m-1}(x) - R_m(x) \quad (m \geq 1),$$

with a separately correct scale-zero term. Since $0 \leq y_m < 2^{-m}$ and $|\mathcal{F}(y)| \leq 2y$ near the origin, $|R_m| \leq 2^{1-m}$.

5.2 The sparse digit form on the unit interval

For $0 \leq x < 1$, write the floor-selected binary expansion

$$x = 0.\delta_1\delta_2\delta_3\cdots = \sum_{m \geq 1} \delta_m 2^{-m}, \quad \delta_m \in \{0, 1\}, \quad (66)$$

where p_m is the integer represented by the first m digits. Let $s_m = \delta_1 + \cdots + \delta_m$ and let

$$t_m = \sum_{\ell > m} \delta_\ell 2^{-\ell} = y_m(x) \quad (67)$$

be the unscaled tail. If $\delta_m = 1$, then $p_m = 2p_{m-1} + 1$ and

$$\alpha_m = (-1)^{s_m} (-1) = (-1)^{s_{m-1}};$$

if $\delta_m = 0$, then $\alpha_m = 0$.

Corollary 5.2 (Digit-sparse series). *For $0 \leq x < 1$,*

$$\boxed{F(x) = \sum_{\substack{m \geq 1 \\ \delta_m = 1}} (-1)^{s_{m-1}} \mathcal{T}_m(t_m).} \quad (68)$$

At a dyadic rational the floor convention chooses the terminating binary expansion, so the sum is finite and contains exactly one term for each set bit of the numerator.

This is the most direct connection between the fast bit-recursive evaluator and the arbitrary argument series: the finite recursion is not a separate algorithm pasted onto the infinite formula; it is literally the same telescope when the binary tail eventually becomes zero.

5.3 Replacing each Taylor coefficient by a finite q -identity

For $Q \in \mathbb{C}$ define the finite inverse-block numerator

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + Q)^{n+k}. \quad (69)$$

The exact one-block q -formula is equivalent to

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n} = \frac{2^n d_n}{n!}, \quad (70)$$

which is independent of Q . Insert these finite identities into the polynomial (59):

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n}. \quad (71)$$

Theorem 5.3 (q -Taylor identity and global q -series). *For every m, y , and $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y). \quad (72)$$

Consequently, for fixed Q and every $x \geq 0$,

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x))}, \quad (73)$$

with absolute convergence.

The key word is *termwise*: every finite polynomial $\mathcal{Q}_m^{(Q)}$ already equals $\mathcal{T}_m$ and is already independent of Q before the outer infinite sum is considered. The q -symbols do not create the convergence of the outer series. They merely provide an alternative finite presentation of each Taylor block.

5.4 The dyadic triangle of finite representations

Suppose $x = m/2^s \in [0, 1]$ is written with a terminating binary expansion. Then three different finite mechanisms evaluate the same rational number:

1. (3) sums the Thue–Morse prefix $0 \leq h < m$ and applies the even-moment correction in one shot;
2. (4) uses one geometric row $0 \leq k \leq s$ and two nested cancellations;
3. (68) walks only through the set bits of m , using inverse-dyadic Taylor data.

The finite sums have no natural term-by-term identification. Their equality is mediated by the master coefficient (14) and by the exact Taylor reduction (60). A fourth finite representation, obtained from the discrete limit, appears in the next section.

6 Discrete q -limits, exact stabilization, and extrapolation

6.1 Shifted splines and the finite row

For $p \geq 1$, $Q \in \mathbb{C}$, and $x \in \mathbb{R}$, define

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r (r - 2^p x + Q)^p. \quad (74)$$

At $Q = 1/2$ this is the centered spline S_p . The formal development proves the exact fixed-cutoff Taylor relation and the uniform bound

$$|S_p^{(Q)}(x) - S_p(x)| \leq 2^{-(p-1)}(e^{|Q-1/2|} - 1). \quad (75)$$

Together with (34),

$$|S_p^{(Q)}(x) - \mathcal{F}(x)| \leq (2e^{|Q-1/2|} - 1)2^{-p}. \quad (76)$$

The discrete q -approximant is

$$\begin{aligned} \mathcal{D}_n^{(Q)}(x) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{N_{n+k}(x)-1} \varepsilon_r(r - 2^{n+k}x + Q)^{n+k}. \end{aligned} \quad (77)$$

For every fixed Q and every real x , the Lean development proves

$$\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x). \quad (78)$$

The notation strongly resembles the exact dyadic formula (4). The following result makes the resemblance literal.

6.2 At dyadic inputs the limit stabilizes exactly

Theorem 6.1 (Finite stabilization on the dyadic grid). *Let $x = m/2^s$ with $m, s \in \mathbb{N}$. Then, for every $n \geq s$ and every $Q \in \mathbb{C}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) = \mathcal{F}(x)}. \quad (79)$$

If $0 \leq x \leq 1$, the right side is $F(x)$.

Proof. For $n \geq s$ put $M = m2^{n-s}$, so that $x = M/2^n$. For every $0 \leq k \leq n$,

$$2^{n+k}x = M2^k \in \mathbb{N}, \quad N_{n+k}(x) = \left\lfloor M2^k + \frac{1}{2} \right\rfloor = M2^k.$$

Therefore (77) becomes exactly the dyadic formula (4) with numerator M and denominator exponent n . That formula is valid for unreduced dyadic representations and for every complex translation Q . $\square$

This is the sharpest connection between the exact and limiting formulae: the discrete formula extends the exact dyadic row to arbitrary x , and on a dyadic grid it needs no limiting process once the row depth reaches the denominator depth.

Corollary 6.2 (Exact finite shifted-spline extrapolation). *For $x = m/2^s$, $n \geq s$, and every $Q \in \mathbb{C}$,*

$$\mathcal{F}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (80)$$

where the weights $\omega_{n,j}$ are defined in (82) below.

Thus every dyadic value admits an exact representation as a finite signed combination of consecutive spline degrees, all evaluated at the same point. Each shifted spline may depend strongly on Q , but the complete combination does not.

6.3 The row polynomial

Reindexing (77) by $j = n - k$ gives the exact identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (81)$$

with

$$\omega_{n,j} = \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-(j+1)}}{(1/2; 1/2)_n}. \quad (82)$$

The full row is encoded by one normalized q -Pochhammer polynomial.

Proposition 6.3 (Generating polynomial of the Toeplitz row). *Define*

$$W_n(z) = \sum_{j=0}^n \omega_{n,j} z^j. \quad (83)$$

Then

$$W_n(z) = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n} = \prod_{\ell=1}^n \frac{1 - z/2^\ell}{1 - 1/2^\ell}. \quad (84)$$

Consequently

$$W_n(1) = 1, \quad (85)$$

$$W_n(2^r) = 0 \quad (1 \leq r \leq n), \quad (86)$$

$$\sum_{j=0}^n |\omega_{n,j}| = W_n(-1) = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \quad (87)$$

$$\sum_{j=0}^n |\omega_{n,j}| 2^j = W_n(-2) = \frac{(-1; 1/2)_n}{(1/2; 1/2)_n}. \quad (88)$$

Proof. Apply the finite q -binomial theorem to $(z/2; 1/2)_n$. Its coefficient of z^j is

$$(-1)^j 2^{-(j)} 2^{-j} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} = (-1)^j 2^{-(j+1)} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2},$$

which gives (84) after division by $(1/2; 1/2)_n$. The evaluations at $1, 2^r, -1, -2$ follow from the product and from the alternating signs of the coefficients. $\square$

The familiar row-sum identity and bounded variation are therefore only two evaluations of a single polynomial. The roots $2, 4, \dots, 2^n$ contain additional information that the basic Toeplitz proof does not need.

6.4 A q -Pochhammer filter in the degree variable

Fix x and Q and regard $p \mapsto S_p^{(Q)}(x)$ as a sequence. Let the backward shift L act by $LS_p^{(Q)} = S_{p-1}^{(Q)}$. Then (81) and (84) can be written as

$$\mathcal{D}_n^{(Q)}(x) = W_n(L) S_{2n}^{(Q)}(x) = \prod_{\ell=1}^n \frac{1 - 2^{-\ell} L}{1 - 2^{-\ell}} S_{2n}^{(Q)}(x). \quad (89)$$

Each factor preserves a constant sequence because its value at $L = 1$ is 1. Meanwhile the zero $W_n(2^r) = 0$ means that it removes a geometric error mode with ratio 2^{-r} in the degree.

Proposition 6.4 (q -Richardson exactness). *Suppose a model sequence has the form*

$$u_p = L + \sum_{r=1}^n a_r 2^{-rp}. \quad (90)$$

Then

$$\sum_{j=0}^n \omega_{n,j} u_{2n-j} = L. \quad (91)$$

Proof. The constant contributes $LW_n(1) = L$. The r th error contributes

$$a_r 2^{-2nr} \sum_{j=0}^n \omega_{n,j} 2^{rj} = a_r 2^{-2nr} W_n(2^r) = 0.$$

□

No claim is needed that the actual spline error has a finite expansion of the form (90). The proposition explains the algebraic design of the row: it is a geometric-grid analogue of Richardson extrapolation, with the normalized q -Pochhammer polynomial as its error-annihilating filter.

6.5 A quantitative 4^{-n} convergence bound

The bounded-variation proof of (78) uses only that all sampled degrees $2n - j$ tend to infinity. Retaining their actual sizes gives a stronger quantitative consequence.

Theorem 6.5 (Explicit discrete-row error). *For $n \geq 1$, $Q \in \mathbb{C}$, and every $x \in \mathbb{R}$, let*

$$H_n = \frac{(-1; 1/2)_n}{(1/2; 1/2)_n}. \quad (92)$$

Then

$$\boxed{|\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| \leq H_n (2e^{|Q-1/2|} - 1) 4^{-n}.} \quad (93)$$

Moreover $H_n \leq 64$, so the convergence is uniformly $\mathcal{O}_Q(4^{-n})$ on the whole real line. At the centered shift,

$$|\mathcal{D}_n^{(1/2)}(x) - \mathcal{F}(x)| \leq H_n 4^{-n}. \quad (94)$$

Proof. Apply (76) with $p = 2n - j$ in (81):

$$\begin{aligned} |\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| &\leq (2e^{|Q-1/2|} - 1) \sum_{j=0}^n |\omega_{n,j}| 2^{-(2n-j)} \\ &= (2e^{|Q-1/2|} - 1) 4^{-n} \sum_{j=0}^n |\omega_{n,j}| 2^j. \end{aligned}$$

Use (88). Finally

$$H_n = \frac{2}{1 - 2^{-n}} \frac{(-1/2; 1/2)_{n-1}}{(1/2; 1/2)_{n-1}} \leq 4 \cdot 16 = 64,$$

where the last factor is the uniform variation bound proved in the formal development. □

The same argument isolates the finite translation dependence.

Corollary 6.6 (Decay of the shift dependence). *For every $x \in \mathbb{R}$ and $n \geq 1$,*

$$|\mathcal{D}_n^{(Q)}(x) - \mathcal{D}_n^{(1/2)}(x)| \leq 2H_n (e^{|Q-1/2|} - 1) 4^{-n}. \quad (95)$$

Thus the finite rows can depend dramatically on Q at small n , but the dependence is uniformly suppressed at a geometric rate. This is analytically different from theorem 4.4, where the finite dyadic expression is exactly constant in Q .

6.6 A synchronized nearest-dyadic q -limit

There is a simpler way to turn the exact dyadic formula into a limit at arbitrary argument. For $x \geq 0$ define the nearest nonnegative dyadic numerator

$$M_n(x) = \max\left(0, \left\lfloor 2^n x + \frac{1}{2} \right\rfloor\right) = N_n(x), \quad \tilde{x}_n = \frac{M_n(x)}{2^n}. \quad (96)$$

Use this one numerator at every inner scale:

$$\begin{aligned} \tilde{\mathcal{D}}_n^{(Q)}(x) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{M_n(x)2^k-1} \varepsilon_r(r - M_n(x)2^k + Q)^{n+k}. \end{aligned} \quad (97)$$

Unlike (77), every finite row is now an exact dyadic formula.

Theorem 6.7 (Synchronized exact-grid representation). *For every $n, Q \in \mathbb{C}$, and $x \geq 0$,*

$$\boxed{\tilde{\mathcal{D}}_n^{(Q)}(x) = \mathcal{F}\left(\frac{M_n(x)}{2^n}\right)}. \quad (98)$$

In particular, the left side is exactly independent of Q . On $0 \leq x \leq 1$,

$$\left| \tilde{\mathcal{D}}_n^{(Q)}(x) - F(x) \right| \leq 2^{-n}, \quad (99)$$

and hence

$$F(x) = \lim_{n \rightarrow \infty} \tilde{\mathcal{D}}_n^{(Q)}(x). \quad (100)$$

Proof. Equation (98) is (4) with numerator $M_n(x)$. For $x \in [0, 1]$, $|\tilde{x}_n - x| \leq 2^{-n-1}$ and the bounded Fabius function is 2-Lipschitz, giving (99). $\square$

The same construction applied to (3) gives a purely moment/Thue–Morse limit:

$$\begin{aligned} F(x) &= \lim_{n \rightarrow \infty} \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{M_n(x)-1} \varepsilon_h \\ &\quad \times \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2M_n(x) - 2h - 1)^{n-2k}, \end{aligned} \quad (101)$$

again with error at most 2^{-n} on $[0, 1]$. This representation contains no q -symbols at all: it is exact dyadic arithmetic followed by nearest-grid convergence.

The synchronized limit is conceptually simple but converges at the basic grid rate $\mathcal{O}(2^{-n})$. The asynchronous row (77) is adapted to the natural spline degree $n+k$ at each scale and, by [theorem 6.5](#), achieves the stronger uniform bound $\mathcal{O}_Q(4^{-n})$. The normalized q -Pochhammer row acts as an acceleration filter.

6.7 What fails away from a dyadic point

Let

$$\theta_n(x) = 2^n x - M_n(x), \quad -\frac{1}{2} \leq \theta_n(x) < \frac{1}{2}. \quad (102)$$

Then for every $k \geq 0$,

$$N_{n+k}(x) = M_n(x)2^k + \left\lfloor 2^k \theta_n(x) + \frac{1}{2} \right\rfloor, \quad (103)$$

and the shifted coordinate in the inner power is

$$r - 2^{n+k}x + Q = r - M_n(x)2^k + (Q - 2^k \theta_n(x)). \quad (104)$$

Thus the source discrete row differs from the synchronized exact dyadic row in exactly two ways:

1. the complete block of length $M_n 2^k$ acquires a boundary defect $\lfloor 2^k \theta_n + 1/2 \rfloor$;
2. the common translation Q is replaced by the scale-dependent translation $Q - 2^k \theta_n$.

At a resolved dyadic point, $\theta_n = 0$, so both defects disappear simultaneously; this is another proof of [theorem 6.1](#). Away from the dyadic grid the shifts vary with k , so the finite common-translation invariance of [theorem 4.4](#) cannot be applied across the row.

6.8 Two infinite constructions that share a limit but not terms

The global binary q -series (73) and the discrete limit (78) both evaluate $\mathcal{F}$, but their relation is not that of an infinite series and its partial sums.

Feature	Binary q -series	Discrete q -limit
Outer index	binary scale m	approximation row n
Finite inner object	$\mathcal{Q}_m^{(Q)} = \mathcal{T}_m$	shifted splines $S_{2n-j}^{(Q)}$
Shift dependence	absent in every summand	generally present in every finite row
Convergence source	exact residual telescope	Toeplitz summability of spline convergence
At dyadic x	outer series terminates at the last set bit	rows stabilize exactly once n reaches the denominator depth
Natural error bound	2^{1-N} after scale N	$H_n(2e^{ Q-1/2 } - 1)4^{-n}$

Table 3: The two arbitrary-argument q constructions.

Warning 6.8. It is therefore incorrect to call $\mathcal{D}_n^{(Q)}$ the n th partial sum of the global q -binomial series. The two families use the same finite half-base identities, but they insert them into different analytic frameworks.

6.9 The first rows

For reference, the first normalized Toeplitz rows are

$$\begin{aligned} (\omega_{1,j})_{j=0}^1 &= (2, -1), \\ (\omega_{2,j})_{j=0}^2 &= (8/3, -2, 1/3), \\ (\omega_{3,j})_{j=0}^3 &= (64/21, -8/3, 2/3, -1/21), \\ (\omega_{4,j})_{j=0}^4 &= (1024/315, -64/21, 8/9, -2/21, 1/315). \end{aligned}$$

Their ordinary and 2^j -weighted variations are shown in [table 4](#).

n	$\sum_j \omega_{n,j} $	$H_n = \sum_j \omega_{n,j} 2^j$
1	3	4
2	5	8
3	45/7	80/7
4	51/7	96/7
5	1683/217	3264/217

Table 4: Two evaluations of the row polynomial: $W_n(-1)$ and $W_n(-2)$.

7 A unified dyadic identity and a worked example

7.1 The complete equality chain

Let $x = m/2^s \in [0, 1]$, use the terminating binary digits $x = 0.\delta_1 \cdots \delta_s$, and let $N \geq s$. Combining the preceding results gives the following chain of exact finite expressions:

$$F(x) = 2^{-\binom{s}{2}} [X^s] (B_m(X) \mathcal{A}(X)) \quad (105)$$

$$= \frac{2^{-\binom{s+1}{2}}}{s!} \sum_{h=0}^{m-1} \varepsilon_h \sum_{k=0}^{\lfloor s/2 \rfloor} \binom{s}{2k} c_k (2m - 2h - 1)^{s-2k} \quad (106)$$

$$= \frac{1}{2^{s^2} (1/2; 1/2)_s} \sum_{k=0}^s \frac{\left[\begin{smallmatrix} s \\ k \end{smallmatrix} \right]_{1/2}}{4^{\binom{k}{2}} (s+k)!} \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + Q)^{s+k} \quad (107)$$

$$= \sum_{k=0}^{\lfloor s/2 \rfloor} \frac{c_k}{(2k)! 2^{k(2s-2k+1)}} S_{s-2k} \left(\frac{m}{2^{s-2k}} \right) \quad (108)$$

$$= \sum_{\substack{1 \leq j \leq s \\ \delta_j = 1}} (-1)^{\delta_1 + \cdots + \delta_{j-1}} \mathcal{T}_j \left(\sum_{\ell > j} \delta_\ell 2^{-\ell} \right) \quad (109)$$

$$= \sum_{j=0}^N \omega_{N,j} S_{2N-j}^{(Q)}(x). \quad (110)$$

The letter k has unrelated local meanings in different lines; this is unavoidable because one line indexes even moments and another indexes geometric scales. The equalities are not termwise transformations. They are different factorizations of the same exact value:

- (105) is the common scalar invariant;
- (106) expands the probability law in centered moments;
- (107) extracts the same coefficient by geometric interpolation;
- (108) corrects one matched spline through even moments;
- (109) telescopes through the set bits;
- (110) extrapolates consecutive spline degrees at the same point.

7.2 The value at 5/16

Take

$$x = \frac{5}{16} = 0.0101_2, \quad F(x) = \frac{305857}{2073600} \approx 0.1475004822530864.$$

This one value illustrates every bridge above.

7.2.1 Master coefficient and moment formula

The prefix polynomial is

$$B_5(X) = e^{4X} - e^{3X} - e^{2X} + e^X - 1. \quad (111)$$

Using $d_0 = 1, d_1 = 1/2, d_2 = 5/18, d_3 = 1/6, d_4 = 143/1350$ in $\mathcal{A}$ gives

$$[X^4](B_5\mathcal{A}) = \frac{305857}{32400}. \quad (112)$$

Since $2^{-\binom{4}{2}} = 1/64$, (14) gives (7.2). Equivalently, with $c_1 = 1/9$ and $c_2 = 19/675$,

$$F(5/16) = \frac{2^{-10}}{4!} \sum_{h=0}^4 \varepsilon_h [t_h^4 + 6c_1 t_h^2 + c_2], \quad t_h = 9 - 2h. \quad (113)$$

7.2.2 Moment correction of the matched spline

The three spline values required by (36) are

$$S_4(5/16) = \frac{1205}{8192}, \quad S_2(5/4) = \frac{15}{16}, \quad S_0(5) = -1. \quad (114)$$

Hence

$$\begin{aligned} F(5/16) &= S_4(5/16) + \frac{1}{2304} S_2(5/4) + \frac{19}{16588800} S_0(5) \\ &= \frac{1205}{8192} + \frac{5}{12288} - \frac{19}{16588800} = \frac{305857}{2073600}. \end{aligned} \quad (115)$$

The matched degree-four spline is already close, but the exact discrepancy is resolved by two universal lower-degree corrections.

7.2.3 The two set bits

The binary digits equal 1 only at positions 2 and 4. The reduction polynomial at the first position is

$$\mathcal{T}_2(y) = \frac{5}{72} + y + 4y^2, \quad (116)$$

and $\mathcal{T}_4(0) = F(1/16) = 143/2073600$. The signs are + at position 2 and – at position 4, so

$$\begin{aligned} F(5/16) &= \mathcal{T}_2(1/16) - \mathcal{T}_4(0) \\ &= \frac{85}{576} - \frac{143}{2073600} = \frac{305857}{2073600}. \end{aligned} \quad (117)$$

The prefix formula sums five Thue–Morse terms; the binary formula needs only the two set bits.

7.2.4 Exact q -extrapolation of centered splines

At row $N = 4$, (110) reads

$$\begin{aligned} F(5/16) &= \frac{1024}{315} S_8(5/16) - \frac{64}{21} S_7(5/16) + \frac{8}{9} S_6(5/16) \\ &\quad - \frac{2}{21} S_5(5/16) + \frac{1}{315} S_4(5/16). \end{aligned} \quad (118)$$

The individual spline values are

$$\begin{aligned} S_4 &= \frac{1205}{8192}, & S_5 &= \frac{289799}{1966080}, & S_6 &= \frac{13917509}{94371840}, \\ S_7 &= \frac{74236247}{503316480}, & S_8 &= \frac{395939357}{2684354560}, \end{aligned} \quad (119)$$

all evaluated at $5/16$. None equals $F(5/16)$, but the q -Pochhammer filter combines them exactly.

7.2.5 Finite shift dependence before stabilization

The first discrete rows illustrate the difference between exact finite translation invariance and asymptotic translation independence.

n	$\mathcal{D}_n^{(0)}(5/16)$	$\mathcal{D}_n^{(1/2)}(5/16)$	$\mathcal{D}_n^{(3)}(5/16)$
1	0.156250000	0.156250000	3.906250000
2	0.151041667	0.147460938	−0.531250000
3	0.147505374	0.147501983	0.125788484
4	0.147500482	0.147500482	0.147500482

Table 5: The row becomes exactly equal to $F(5/16)$ at $n = 4$, independently of Q .

The exact dyadic q -formula itself is independent of Q already at row four, although its outer summands vary substantially. Their decimal contributions are:

scale k	$Q = 0$	$Q = 1/2$	$Q = 3$
0	0.000626240	0.000466967	0.000000000
1	−0.016345021	−0.014038037	−0.005184307
2	0.141657926	0.131089095	0.084488329
3	−0.467452954	−0.449506045	−0.365062120
4	0.489014292	0.479488502	0.433258580
sum	0.147500482	0.147500482	0.147500482

Table 6: Different translated decompositions of the same exact dyadic value.

8 Computational and conceptual consequences

8.1 Which representation is best for which task?

The formulae are mathematically equivalent at dyadic inputs, but not computationally interchangeable.

The exact q -formula contains inner blocks of length $m2^k$, so a direct implementation can be expensive. Its value is primarily structural: it exposes geometric interpolation, translation invariance, and the extrapolation filter. For exact numerical evaluation, the bit recursion or the master coefficient with precomputed moments is generally preferable.

8.2 A compact exact-arithmetic scheme

Equation (31) computes all half moments with rational arithmetic, after which (18) evaluates a dyadic value.

Algorithm 8.1 (Moment-prefix evaluator). Given $m, n \in \mathbb{N}$:

1. Set $d_0 = 1$ and, for $1 \leq j \leq n$, compute

$$d_j = \frac{1}{2^j - 1} \sum_{r=1}^j \binom{j}{r} \frac{d_{j-r}}{r+1}.$$

2. Form

$$V = \sum_{h=0}^{m-1} \varepsilon_h \sum_{j=0}^n \binom{n}{j} d_j (m-1-h)^{n-j}.$$

Task	Recommended representation	Reason
Prove rationality or denominator facts	master coefficient or moment formula	all arithmetic is finite and the universal moments are explicit rationals
Evaluate one dyadic with a sparse numerator	binary Taylor reduction	the number of reductions is the number of set bits
Evaluate many numerators at one depth	precompute d_j or c_k , then use prefix recurrences	universal moment data are reused
Explain the q -coefficients	geometric divided difference and residue form	Gaussian coefficients are recognized as interpolation weights, not mysterious special-function decorations
Approximate arbitrary x with a simple certified error	centered spline S_p	direct global error 2^{-p} and one finite Thue–Morse sum
Accelerate arbitrary-argument spline convergence	discrete row $\mathcal{D}_n^{(Q)}$	exact q -Pochhammer filter and uniform $\mathcal{O}_Q(4^{-n})$ bound
Use only exact dyadic arithmetic in a limiting scheme	synchronized row $\tilde{\mathcal{D}}_n^{(Q)}$ or synchronized moment limit	every finite stage is an exact dyadic value and is translation-independent
Trace dependence on binary digits	digit-sparse series	one nonzero summand per exposed 1-bit digits

Table 7: Representation selection. The q -formula is structurally illuminating but is usually not the cheapest naive exact evaluator.

3. Return $2^{-\binom{n}{2}}V/n!$.

The result is $\mathcal{F}(m/2^n)$ and is $F(m/2^n)$ when $0 \leq m \leq 2^n$.

A literal pseudocode version is included for clarity.

Listing 1: Exact dyadic evaluation from half moments.

```

d[0] := 1
for j := 1, ..., n do
  s := 0
  for r := 1, ..., j do
    s := s + binomial(j, r) * d[j-r] / (r+1)
  end
  d[j] := s / (2^j - 1)
end

V := 0
for h := 0, ..., m-1 do
  inner := 0
  for j := 0, ..., n do
    inner := inner + binomial(n, j) * d[j] * (m-1-h)^(n-j)
  end
  V := V + (-1)^(binary_weight(h)) * inner
end
return V / (2^(n*(n-1)/2) * n!)

```

The centered-moment formula can be evaluated similarly. Its advantage is parity: only $\lfloor n/2 \rfloor + 1$ moments occur. The half-moment formula aligns more directly with the binary Taylor polynomials.

8.3 Exact arithmetic checks implied by the connections

The network of representations supplies a strong verification suite. For chosen m, n one can independently check:

1. the master coefficient against the centered-moment formula;
2. the half-base q expression at two or more translations Q ;
3. the terminating binary reduction when $m/2^n \in [0, 1]$;
4. the moment-corrected matched-spline identity;
5. the exact q -extrapolation row at any order $N \geq n$;
6. representation independence under $(m, n) \mapsto (2m, n + 1)$.

These checks stress different cancellation patterns. Agreement is therefore more informative than recomputing the same formula in a rearranged order. The last item is now more than a check: `qBinomialThueMorseDyadicFormula_refine` proves it exactly. The stronger `qBinomialThueMorseDyadicTranslatedFormula_eq_of_rat_eq` allows any two numerator–exponent pairs representing the same rational dyadic value and permits different rational translations on the two sides. The translated refinement and half-shifted variants are proved in `FabiusQBinomialFormula.lean`. These theorem-level invariances do not formalize the separate finite-row stabilization question in [section 6](#).

8.4 Endpoint and convention warnings

Several small conventions are mathematically active.

Warning 8.2 (The scale-zero term). The global binary series begins at $m = 0$. It happens to vanish for $0 \leq x < 1$, but it equals 1 at $x = 1$ and controls the signed behavior on integer blocks. A series beginning at $m = 1$ is valid only on the half-open unit interval.

Warning 8.3 (Integer exponents). The exponent $\binom{j+1}{2} - \binom{m-j}{2}$ in (59) is an integer and may be negative. Replacing integer subtraction by truncated natural subtraction changes the polynomial.

Warning 8.4 (Centered powers are signed). Expressions such as $(r - m2^k + Q)^{n+k}$ live in a characteristic-zero scalar field. The quantity $r - m2^k$ is deliberately negative on much of the block. Truncated subtraction in $\mathbb{N}$ destroys the formula.

Warning 8.5 (Binary expansions at dyadics). The floor prefixes $p_m = \lfloor 2^m x \rfloor$ select the terminating expansion at a dyadic rational. Replacing it by the alternative expansion ending in infinitely many 1-digits changes the termwise digit series even though the represented real number is unchanged.

9 Relation to the Lean development

The principal source theorems are formalized in the following modules. Filenames are relative to `Analysis/FabiusFunction`.

Module	Role in the present article
<code>DyadicClosedForm.lean</code>	Exact Thue–Morse/even-moment formula for arbitrary dyadic numerators.
<code>HalfQBinomial.lean</code>	Finite q -Pochhammer and Gaussian-binomial calculus at $q = 1/2$, including the interpolation annihilation row.
<code>ThueMorseExponential.lean</code>	Complete-block exponential factorization, Prouhet cancellation, and translation of inner powers.
<code>FabiusQBinomialFormula.lean</code>	Centered finite q -binomial formula, arbitrary numerators, common-translation invariance, and invariance under equal rational dyadic representations.
<code>FabiusQBinomialTaylor.lean</code>	Identification of the finite q expression with the Taylor reduction polynomial.

Module	Role in the present article
<code>FabiusQBinomialScalarFormula.lean</code> and <code>FabiusDyadicQBinomialScalar.lean</code>	Scalar normalizations, real/complex translated forms, and the constant-polynomial coefficient theorem.
<code>FabiusRawQBinomialFormula.lean</code> and <code>FabiusRawQBinomialScalar.lean</code>	Raw inverse-dyadic forms and their scalar versions.
<code>FabiusBinaryReductionSeries.lean</code>	Exact binary residual telescope, absolute convergence, and uniform tail bound.
<code>FabiusParityPowerSeries.lean</code>	Corrected scale-zero parity-power presentation.
<code>FabiusGlobalQBinomialSeries.lean</code>	Termwise substitution of finite q -identities into the binary telescope.
<code>FabiusComplexShiftSpline.lean</code>	Shifted finite splines, fixed-cutoff Taylor expansion, and uniform shift bound.
<code>FabiusDiscreteLimitToeplitz.lean</code>	Row mass, exact total variation, uniform variation bound, and finite-row Toeplitz theorem.
<code>FabiusDiscreteLimitIntegration.lean</code>	Reindexing of the discrete formula and convergence to the signed global function.

The following statements are direct paper-level corollaries or reorganizations of those formalized results:

- the expectation and two contour representations (15), (17), and (51);
- the composition and Bell-polynomial formulae for d_n ;
- the moment-corrected spline transform and its reciprocal (36)–(41);
- exact finite stabilization of the discrete row at dyadic arguments;
- the row polynomial W_n , the shift-operator form, and the q -Richardson interpretation;
- the explicit $\mathcal{O}_Q(4^{-n})$ bound (93);
- the synchronized nearest-dyadic q - and moment-limits.

They require no new unproved property of the Fabius function; each follows by finite algebra, coefficient extraction, or the already formalized uniform spline estimates. They are marked here to distinguish conceptual reorganization from theorem names that already occur verbatim in the repository.

10 Conclusion

The exact dyadic, global-series, and discrete-limit formulae are best understood not as a collection of special tricks but as a small number of reusable operators acting in different variables.

The product $B_m \mathcal{A}$ separates numerator arithmetic from the universal probability law. Expanding its coefficient in centered moments gives the classical Thue–Morse formula; reading it as an expectation gives a probabilistic representation; taking a Cauchy integral gives an analytic one. Introducing the geometric node z turns the same coefficient into the leading term of a degree- n polynomial. The Gaussian row at base $1/2$ is then simply the barycentric divided difference on $-1, -2, \dots, -2^n$, and the q -Pochhammer symbol is the node polynomial of that grid.

The centered finite spline sits directly inside the exact arithmetic. Its matched-grid value is the zeroth centered-moment term, and the remaining moments form an exact, invertible defect correction. A second correction uses the normalized q -Pochhammer polynomial in the spline degree: it preserves constants, annihilates geometric error modes, and yields exact recovery at dyadic points. The resulting

discrete row is therefore naturally viewed as a q -Richardson extrapolator, not merely as an opaque Toeplitz average.

Finally, the two arbitrary-argument q constructions have distinct logical roles. The global q -series is the binary residual telescope with an alternative finite formula inserted into each summand. The discrete limit is a finite extrapolation row whose shift dependence disappears only asymptotically. Their meeting point is the dyadic grid: there the binary series terminates, the discrete row stabilizes, the synchronized row is exact, and all roads return to the same master coefficient.

A A compact identity sheet

For convenience, the principal transformations are collected here.

Probability and moments

$$\begin{aligned}
 Y &= \sum_{j \geq 1} U_j / 2^j, & Z &= 2Y - 1, \\
 \mathcal{A}(X) &= \mathbb{E}(e^{XY}) = \prod_{j \geq 1} \frac{e^{X/2^j} - 1}{X/2^j}, \\
 \mathcal{A}(X) &= e^{X/2} \mathcal{C}(X/2), & \mathcal{C}(T) &= \sum_{k \geq 0} c_k T^{2k} / (2k)!, \\
 d_n &= 2^{-n} \sum_{k \leq n/2} \binom{n}{2k} c_k, \\
 c_k &= \sum_{j=0}^{2k} (-1)^j 2^j \binom{2k}{j} d_j.
 \end{aligned}$$

Master coefficient

$$\begin{aligned}
 B_m(X) &= \sum_{h < m} \varepsilon_h e^{(m-1-h)X}, \\
 \mathcal{F}(m/2^n) &= 2^{-\binom{n}{2}} [X^n](B_m \mathcal{A}), \\
 \mathcal{F}(m/2^n) &= \frac{2^{-\binom{n}{2}}}{n!} \sum_{h < m} \varepsilon_h \mathbb{E}(m-1-h+Y)^n.
 \end{aligned}$$

Geometric extraction

$$\begin{aligned}
 w_{n,k} &= (-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2}, & x_k &= -2^k, \\
 \frac{1}{\mathfrak{m}_n} \sum_{k=0}^n w_{n,k} P(x_k) &= [z^n] P(z) \quad (\deg P \leq n), \\
 \mathfrak{m}_n &= 2^{\binom{n+1}{2}} (1/2; 1/2)_n.
 \end{aligned}$$

Spline corrections

$$\begin{aligned}\mathcal{F}(m/2^n) &= \sum_{k \leq n/2} \frac{c_k}{(2k)!2^{k(2n-2k+1)}} S_{n-2k}(m/2^{n-2k}), \\ S_n(m/2^n) &= \sum_{k \leq n/2} \frac{\gamma_k}{(2k)!2^{k(2n-2k+1)}} \mathcal{F}(m/2^{n-2k}), \\ \sum_{k \geq 0} \gamma_k T^{2k} / (2k)! &= 1/\mathcal{C}(T).\end{aligned}$$

Discrete filter

$$\begin{aligned}\mathcal{D}_n^{(Q)}(x) &= \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \\ W_n(z) &= \sum_j \omega_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}, \\ |\mathcal{D}_n^{(Q)}(x) - \mathcal{F}(x)| &\leq \frac{(-1; 1/2)_n}{(1/2; 1/2)_n} (2e^{|Q-1/2|} - 1) 4^{-n}, \\ x = m/2^s, \ n \geq s &\implies \mathcal{D}_n^{(Q)}(x) = \mathcal{F}(x).\end{aligned}$$

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